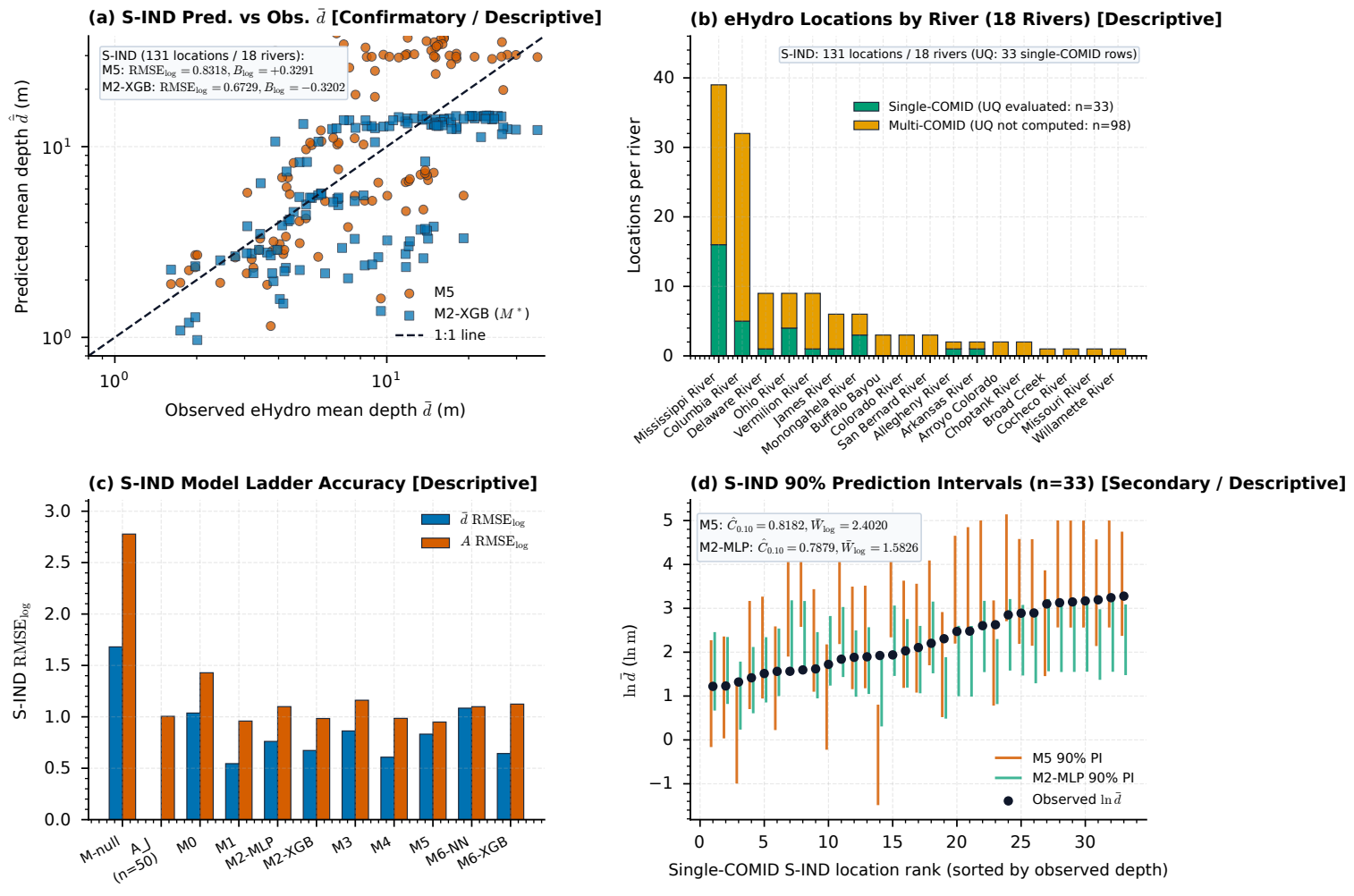

(a) S-IND Pred. vs Obs. $\bar{d}$ [Confirmatory / Descriptive]
S-IND (131 locations / 18 rivers):
M5: $\mathrm{RMSE}_{\log} = 0.8318$, $B_{\log} = +0.3291$
M2-XGB: $\mathrm{RMSE}_{\log} = 0.6729$, $B_{\log} = -0.3202$
M5
M2-XGB ($M^*$)
1:1 line
Predicted mean depth $\hat{\bar{d}}$ (m)
Observed eHydro mean depth $\bar{d}$ (m)
(b) eHydro Locations by River (18 Rivers) [Descriptive]
S-IND: 131 locations / 18 rivers (UQ: 33 single-COMID rows)
Single-COMID (UQ evaluated: n=33)
Multi-COMID (UQ not computed: n=98)
Locations per river
Mississippi River
Columbia River
Delaware River
Ohio River
Vermilion River
James River
Monongahela River
Buffalo Bayou
Colorado River
San Bernard River
Allegheny River
Arkansas River
Arroyo Colorado
Choptank River
Broad Creek
Cocheco River
Missouri River
Willamette River
(c) S-IND Model Ladder Accuracy [Descriptive]
$\bar{d}$ $\mathrm{RMSE}_{\log}$
$A$ $\mathrm{RMSE}_{\log}$
S-IND $\mathrm{RMSE}_{\log}$
M-null
A_J (n=50)
M0
M1
M2-MLP
M2-XGB
M3
M4
M5
M6-NN
M6-XGB
(d) S-IND 90% Prediction Intervals (n=33) [Secondary / Descriptive]
M5: $\hat{C}_{0.10} = 0.8182$, $\bar{W}_{\log} = 2.4020$
M2-MLP: $\hat{C}_{0.10} = 0.7879$, $\bar{W}_{\log} = 1.5826$
M5 90% PI
M2-MLP 90% PI
Observed $\ln \bar{d}$
$\ln \bar{d}$ (ln m)
Single-COMID S-IND location rank (sorted by observed depth)